

First-Order Logic in Artificial intelligence

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- In the topic of Propositional logic, we have seen that how to represent statements using propositional logic.
- But unfortunately, in propositional logic, we can only represent the facts, which are either true or false. PL is not sufficient to represent the complex sentences or natural language statements.
- The propositional logic has very limited expressive power. Consider the following sentence, which we cannot represent using PL logic.

First-Order Logic in Artificial intelligence

- "Some humans are intelligent", or
- "Sachin likes cricket."
- To represent the above statements, PL logic is not sufficient, so we required some more powerful logic, such as first-order logic.

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First-Order logic:

- First-order logic is another way of knowledge representation in artificial intelligence. It is an extension to propositional logic.
- FOL is sufficiently expressive to represent the natural language statements in a concise way.
- First-order logic is also known as **Predicate logic or First-order predicate logic**. First-order logic is a powerful language that develops information about the objects in a more easy way and can also express the relationship between those objects.

First-Order logic:

- First-order logic (like natural language) does not only assume that the world contains facts like propositional logic but also assumes the following things in the world:
 - **Objects:** A, B, people, numbers, colors, wars, theories, squares, pits, wumpus,
 - **Relations:** It can be **unary relation such as:** red, round, is adjacent, **or n-any relation such as:** the sister of, brother of, has color, comes between
 - **Function:** Father of, best friend, third inning of, end of,

First-Order logic:

- As a natural language, first-order logic also has two main parts:
 - a) **Syntax**
 - b) **Semantics**

Syntax of First-Order logic:

- The syntax of FOL determines which collection of symbols is a logical expression in first-order logic.
- The basic syntactic elements of first-order logic are symbols. We write statements in short-hand notation in FOL.

Basic Elements of First-OrderLogic

- Following are the basic elements of FOL syntax:

Constant	1, 2, A, John, Mumbai, cat,....
Variables	x, y, z, a, b,....
Predicates	Brother, Father, >,....
Function	sqrt, LeftLegOf,
Connectives	$\wedge$, $\vee$, $\neg$, $\Rightarrow$, $\Leftrightarrow$
Equality	$=$
Quantifier	$\forall$, $\exists$

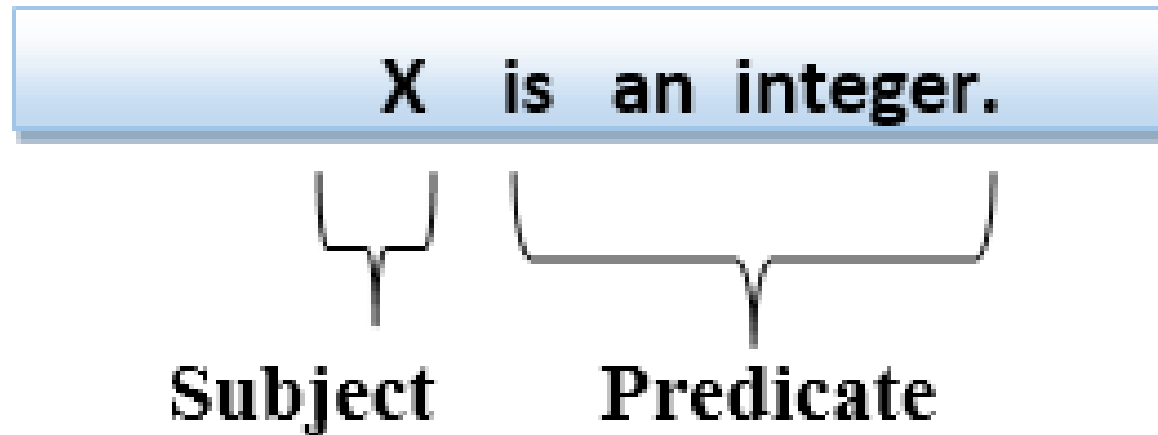
Atomic sentences:

- Atomic sentences are the most basic sentences of first-order logic. These sentences are formed from a predicate symbol followed by a parenthesis with a sequence of terms.
- We can represent atomic sentences as **Predicate (term1, term2,, term n)**.
- **Example: Ravi and Ajay are brothers: => Brothers(Ravi, Ajay).**
Chinky is a cat: => cat (Chinky).

Complex Sentences:

- Complex sentences are made by combining atomic sentences using connectives.
- **First-order logic statements can be divided into two parts:**
 - **Subject:** Subject is the main part of the statement.
 - **Predicate:** A predicate can be defined as a relation, which binds two atoms together in a statement.
- **Consider the statement: "x is an integer."**, it consists of two parts, the first part x is the subject of the statement and second part "is an integer," is known as a predicate.

Complex Sentences:



Quantifiers in First-order logic:

- A quantifier is a language element which generates quantification, and quantification specifies the quantity of specimen in the universe of discourse.
- These are the symbols that permit to determine or identify the range and scope of the variable in the logical expression. There are two types of quantifier:
 - **Universal Quantifier, (for all, everyone, everything)**
 - **Existential quantifier, (for some, at least one).**

Universal Quantifier:

- Universal quantifier is a symbol of logical representation, which specifies that the statement within its range is true for everything or every instance of a particular thing.
- The Universal quantifier is represented by a symbol $\forall$, which resembles an inverted A.
- **Note:** In universal quantifier we use implication " $\rightarrow$ ".

Universal Quantifier:

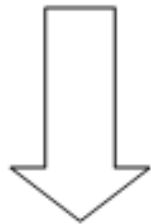
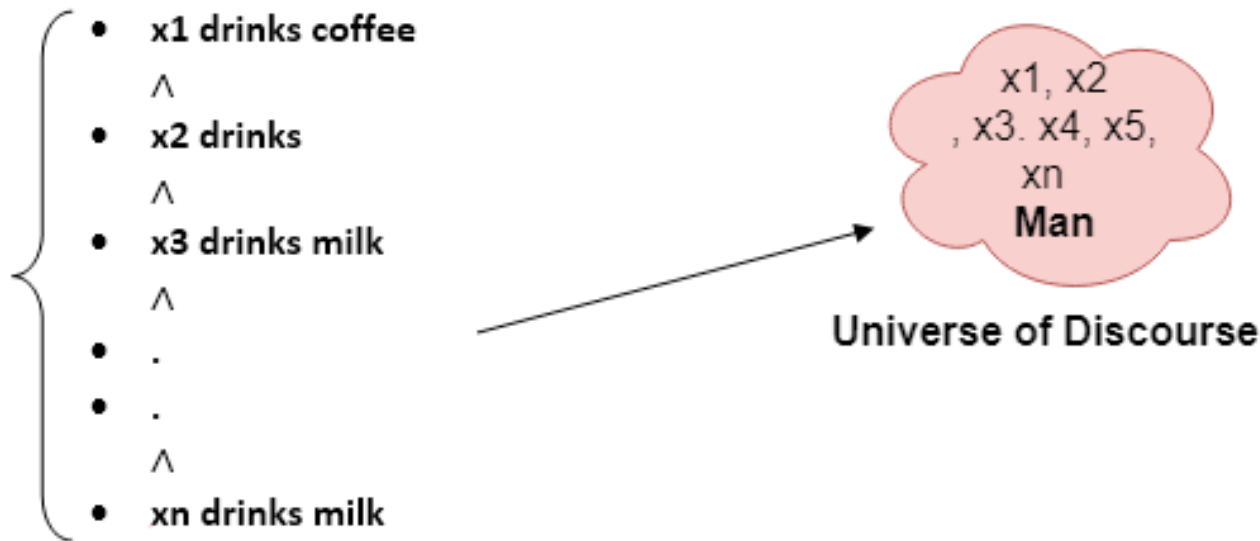
If x is a variable, then $\forall x$ is read as:

- **For all x**
- **For each x**
- **For every x .**

Example:

All man drink coffee.

Let a variable x which refers to a cat so all x can be represented in UOD as below:



So in shorthand notation, we can write it as : **$\forall x \text{ man}(x) \rightarrow \text{drink}(x, \text{coffee}).$**

It will be read as: There are all x where x is a man who drink coffee.

Existential Quantifier:

- Existential quantifiers are the type of quantifiers, which express that the statement within its scope is true for at least one instance of something.
- It is denoted by the logical operator $\exists$, which resembles as inverted E. When it is used with a predicate variable then it is called as an existential quantifier.

Existential Quantifier:

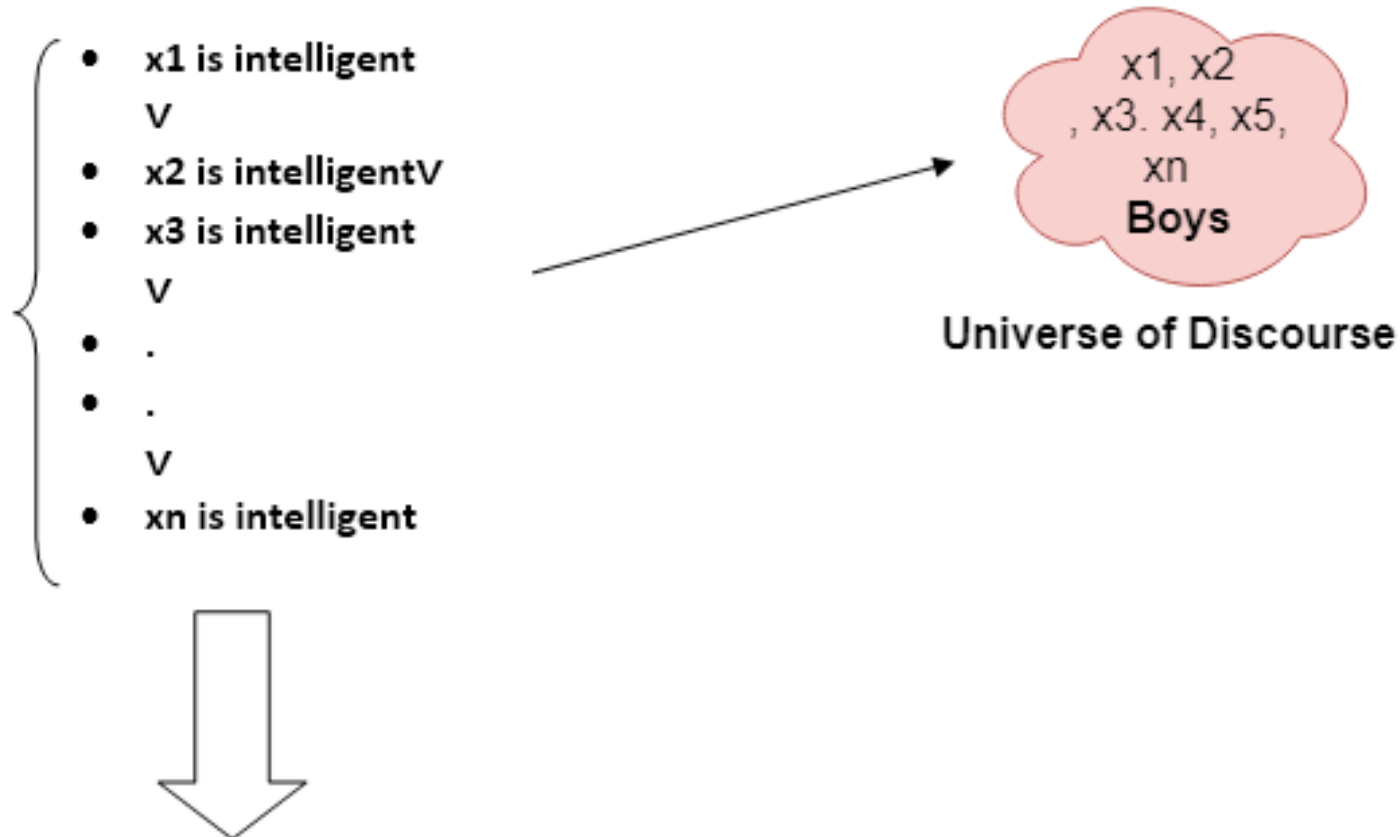
- Note: In Existential quantifier we always use AND or Conjunction symbol ($\wedge$).

If x is a variable, then existential quantifier will be $\exists x$ or $\exists(x)$.
And it will be read as:

- **There exists a 'x.'**
- **For some 'x.'**
- **For at least one 'x.'**

Example:

- Some boys are intelligent.



So in short-hand notation, we can write it as:

$\exists x: \text{boys}(x) \wedge \text{intelligent}(x)$

It will be read as: There are some x where x is a boy who is intelligent.

Points to remember:

- The main connective for universal quantifier $\forall$ is implication $\rightarrow$.
- The main connective for existential quantifier $\exists$ is and $\wedge$.

Properties of Quantifiers:

- In universal quantifier, $\forall x \forall y$ is similar to $\forall y \forall x$.
- In Existential quantifier, $\exists x \exists y$ is similar to $\exists y \exists x$.
- $\exists x \forall y$ is not similar to $\forall y \exists x$.

Some Examples of FOL using quantifier:

1. All birds fly.

In this question the predicate is "fly(bird)."

And since there are all birds who fly so it will be represented as follows.

$$\forall x \text{ bird}(x) \rightarrow \text{fly}(x).$$

2. Every man respects his parent.

In this question, the predicate is "respect(x, y)," where x=man, and y= parent.

Since there is every man so will use $\forall$, and it will be represented as follows:

$$\forall x \text{ man}(x) \rightarrow \text{respects}(x, \text{parent}).$$

Some Examples of FOL using quantifier:

3. Some boys play cricket.

In this question, the predicate is "**play(x, y)**," where x= boys, and y= game. Since there are some boys so we will use **$\exists$** , and it will be represented as:

$\exists x \text{ boys}(x) \rightarrow \text{play}(x, \text{cricket}).$ (wrong)

4. Not all students like both Mathematics and Science.

In this question, the predicate is "**like(x, y)**," where x= student, and y= subject.

Since there are not all students, so we will use **$\forall$** with negation, so following representation for this:

$\neg \forall (x) [\text{student}(x) \rightarrow \text{like}(x, \text{Mathematics}) \wedge \text{like}(x, \text{Science})].$

Some Examples of FOL using quantifier:

5. Everyone is loyal to someone

$$\forall x \exists y : \text{loyal}(x,y)$$

6. Everyone loves everyone

$$\forall x \forall y : \text{loves}(x,y)$$

7. Everyone loves everyone except himself

$$\forall x \forall y : \text{loves}(x,y) \wedge \neg \text{loves}(x,x)$$

8. All Romans were either loyal to Ceasar or hated him

$$\forall x : \text{Roman}(x) \Rightarrow \text{Loyal}(x, \text{Ceasar}) \vee \text{Hated}(x, \text{Ceasar})$$

Some Examples of FOL using quantifier:

9. People only try to assassinate rulers they are not loyal too.

Wrong: $\forall x \forall y : \text{Person}(x) \wedge \text{Ruler}(y) \wedge \neg \text{loyal}(x,y)$
 $\Rightarrow \text{tryassasinate}(x,y)$

Correct: $\forall x \forall y : \text{Person}(x) \wedge \text{Ruler}(y) \wedge \text{tryassasinate}(x,y)$
 $\Rightarrow \neg \text{loyal}(x,y)$

Some Examples of FOL using quantifier:

10 Anyone who is married and has more than one spouse is a bigamist

$$\forall x : \text{Married}(x) \wedge (\text{no.of.spouse}(x) > 1) \\ \Rightarrow \text{bigamist}(x)$$

$$\forall x : \text{Married}(x) \wedge \text{gt}(\text{no.of.spouse}(x), 1) \\ \Rightarrow \text{bigamist}(x)$$

system will process second one easily because we use gt function. In first solution we hard code.

Some Examples of FOL using quantifier:

11. You can fool all the people some of the time

$$\forall x \exists t: \text{Person}(x) \Rightarrow \text{time}(t) \wedge \text{canfool}(x, t)$$

12. You can fool some of the people all the time

$$\exists x \forall t: \text{Person}(x) \wedge \text{time}(t) \Rightarrow \text{canfool}(x, t)$$

Some Examples of FOL using quantifier:

13. Every city has a dog catcher who has been bitten by every dog in the town

$$\forall x \exists y \forall z : \text{City}(x) \wedge \text{Dogcatcher}(y) \wedge \text{Dog}(z) \wedge \text{lives}(z,x) \Rightarrow \text{bites}(z,y)$$

$$\forall x \exists y \forall z : \text{City}(x) \Rightarrow \text{Dogcatcher}(y) \wedge \text{Dog}(z) \wedge \text{lives}(z,x) \wedge \text{bites}(z,y)$$

Pick your answer
(solution on the next
slide)

Some Examples of FOL using quantifier:

13. Every city has a dog catcher who has been bitten by every dog in the town

Solution

$$\forall x \exists y \forall z : \text{City}(x) \Rightarrow \text{Dogcatcher}(y) \wedge \text{Dog}(z) \wedge \text{lives}(z,x) \wedge \text{bites}(z,y)$$

Some Examples of FOL using quantifier:

14. one's husband is one's male spouse

Solution

$$\forall w \exists h : \text{Husband}(h,w) \Rightarrow \text{Male}(h) \wedge \text{Spouse}(h,w)$$

Some Examples of FOL using quantifier:

15 parents and children are inverse relation

$$\forall p \forall c: \text{Parent}(p,c) \Rightarrow \text{Child}(c,p)$$

16 A sibling is another child of ones parent

$$\forall x \forall y \exists p: \text{Sibling}(x,y) \Rightarrow x \neq y \wedge \text{parent}(p,x) \wedge \text{parent}(p,y)$$

Some Examples of FOL using quantifier:

**17. The best score in Maths is always higher than the
best score in English**

$$\forall x \forall y: \text{bestmathscore}(x) \wedge \text{bestenglishscore}(y) \Rightarrow x > y$$

Some Examples of FOL using quantifier:

18. There is a barber who shaves all men in town who do not shave themselves

$$\forall x \forall y \exists z : \text{Man}(x) \wedge \text{Barber}(z) \wedge \text{Town}(y) \wedge \text{Lives}(x,y) \Rightarrow \neg \text{Shaves}(x,x) \wedge \text{Shaves}(z,x)$$

$$\forall x \forall y \exists z : \text{Man}(x) \wedge \text{Town}(y) \wedge \text{Lives}(x,y) \wedge \neg \text{Shaves}(x,x) \Rightarrow \text{Barber}(z) \wedge \text{Shaves}(z,x)$$

$$\forall x \forall y \exists z : \text{Barber}(z) \Rightarrow \text{Man}(x) \wedge \text{Town}(y) \wedge \text{Lives}(x,y) \wedge \neg \text{Shaves}(x,x) \wedge \text{Shaves}(z,x)$$

Think and pick your solution

Some Examples of FOL using quantifier:

18. There is a barber who shaves all men in town who do not shave themselves

Solution

$$\forall x \forall y \exists z : \text{Man}(x) \wedge \text{Town}(y) \wedge \text{Lives}(x,y) \wedge \\ \neg \text{Shaves}(x,x) \Rightarrow \text{Barber}(z) \wedge \text{Shaves}(z,x)$$

Nesting Quantifiers

- Everyone likes some kind of food
 $\forall y \exists x, \text{food}(x) \wedge \text{likes}(y, x)$
- There is a kind of food that everyone likes
 $\exists x \forall y, \text{food}(x) \wedge \text{likes}(y, x)$
- Someone likes all kinds of food
 $\exists y \forall x, \text{food}(x) \wedge \text{likes}(y, x)$
- Every food has someone who likes it
 $\forall x \exists y, \text{food}(x) \wedge \text{likes}(y, x)$

Nesting Quantifiers

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- There is a kind of food that everyone likes
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Quiz

Every gardener likes the sun.

$\forall x \text{ gardener}(x) \rightarrow \text{likes}(x, \text{Sun})$

All purple Mushrooms are poisonous

$\forall x : \text{Mushroom}(x) \wedge \text{purple}(x) \Rightarrow \text{poisonous}(x)$

Every food has someone who likes it

$\forall x \exists y, \text{ food}(x) \wedge \text{likes}(y, x)$

Free and Bound Variables

- The quantifiers interact with variables which appear in a suitable way. There are two types of variables in First-order logic which are given below:
 - **Free Variable:** A variable is said to be a free variable in a formula if it occurs outside the scope of the quantifier.
Example: $\forall x \exists (y)[P(x, y, z)]$, where z is a free variable.
 - **Bound Variable:** A variable is said to be a bound variable in a formula if it occurs within the scope of the quantifier.
Example: $\forall x [A(x) B(y)]$, here x and y are the bound variables.